

10/9/25

Task 6 - Utilizing 'functions' concept in Python Programming

Aim: To write the python program using 'functions' concepts in Python Programming

6.1 - You are developing a small Python script to analyze and manipulate a list of student grades for a class project. Write a python program that satisfy the above requirements using the built-in functions `print()`, `len()`, `type()`, `max()`, `min()`, `sorted()`, `reversed()` and `range()`.

Algorithm:

1. Start the Program

2. Print a welcome message. Outputs a simple greeting

3. Determine and print the number of students:

Uses `len()` to find the number of elements in the `students_names` list.

4. Print the type of lists: Uses `type()` to show the type of the `student_names` and `student_grades` lists

5. Find and print highest and lowest grades. Use `max()` and `min()` to determine the highest and lowest values in `students_grades`

6. Print sorted list of grades: Uses `sorted()` to sort the grades

7. Print reversed list of grades: Uses `reversed()` to reverse the sorted list and converts it to a list

8. Iterate and print a range of grade indices: Uses `range()` to create a list of indices from 1 to the number of students

Output:

Welcome to the Student Grades Analyzer:

Number of Students: 4

Type of student-names list: <class 'list'>

Type of student-grades list: <class 'list'>

Highest grade: 92

Lowest grade: 78

Sorted grades: [78, 85, 90, 92]

Reversed grades: [92, 90, 85, 78]

Grade indices from 1 to number of students: [1, 2, 3, 4]

Program:

```
def analyze_student_grades():  
    student_names = ["Alice", "Bob", "Diana", "Charlie"]  
    student_grades = [85, 92, 90, 78]  
    print("Welcome to the Students Grades Analyzer!\n")  
    num_students = len(student_names)  
    print("Number of Students:", num_students)  
    print("\nType of student_names list:", type(student_names))  
    print("Type of student_grades list:", type(student_grades))  
    highest_grade = max(student_grades)  
    lowest_grade = min(student_grades)  
    print("\nHighest grade:", highest_grade)  
    print("\nLowest grade:", lowest_grade)  
    sorted_grades = sorted(student_grades)  
    print("\nSorted grades:", sorted_grades)  
    reversed_grades = list(reversed(sorted_grades))  
    print("Reversed grades:", reversed_grades)  
    grade_indices = list(range(1, num_students + 1))  
    print("\nGrade indices from 1 to number of students:  
        grade_indices)  
analyze_student_grades()
```

6.2 - Small calculator application to perform basic arithmetic operations (addition, subtraction, multiplication, division)

Algorithm:

1. Start the program
2. User Input for Numbers: The program prompts the user to enter two numbers
3. User Input for Operation: The program prompts the user to choose an arithmetic operation
4. Perform Operation: Based on the user's choice, the program performs the chosen arithmetic operation using the defined functions
5. Display Result: The program displays the result of the operation
6. Stop

Program:

```
def add(a,b):
```

```
    return a+b
```

```
def subtract(a,b):
```

```
    return a-b
```

```
def multiply(a,b):
```

```
    return a*b
```

```
def divide(a,b):
```

```
    if b!=0:
```

```
        return a/b
```

```
    else:
```

```
        return "Error: Division by zero"
```

```
def greet(name):
```

```
    return ("Hello {name} ! Welcome to the program.")
```


Output:

Arithmetic Operations:

Sum of 10 and 5: 15

Difference between 10 and 5: 5

Product of 10 and 5: 50

Quotient of 10 and 5: 2.0

Greeting:

Hello, Alice! Welcome to the program

```

def main():
    num1 = 10
    num2 = 5
    print("Arithmetic Operations!")
    print(f"Sum of {num1} and {num2}:", add(num1, num2))
    print(f"Difference between {num1} and {num2}:", subtract
          (num1, num2))
    print(f"Product of {num1} and {num2}:", multiply(num1,
          num2))
    print(f"Quotient of {num1} and {num2}:", divide(num1, num2))

    user_name = "Alice"
    print("\n Greeting:")
    print(greet(user_name))

if __name__ == "__main__":
    main()

```

VEL TECH	
EX NO.	16
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	15

Result:

Thus the python program using 'Functions' concepts was successfully executed and the output was verified.